Ella Cohen

(703) 888-7921 | efc21@duke.edu | www.linkedin.com/in/ella-cohen-91294220b

EDUCATION

Duke University, Durham, NC **May 2028**

BSE Electrical & Computer Engineering | BS Computer Science

GPA: 3.981 | **Concentration:** Machine Learning

Relevant coursework: Data Structures & Algorithms, Computer Architecture, Signals & Systems, Linear Algebra, Probability, Microelectronic Devices & Circuits

Affiliations: Duke Engineers for International Development, Society of Women Engineers, Duke Technology Scholars (DTech Scholar), Duke Rock Climbing Club (Operations Manager)

TECHNICAL SKILLS

Languages: Python, Java, C, C++, MATLAB, MIPS Assembly, JavaScript, TypeScript
Frameworks & Libraries: Django, NumPy, Matplotlib, JUnit, Tkinter
Tools: Git, Linux/Unix, MySQL, VS Code, Vim, AutoCAD, Onshape, Microsoft Power Automate
Concepts: Data Structures & Algorithms, Object-Oriented Programming, Backend Development, Software Testing & Debugging, Machine Learning, Simulation, Signal Processing

EXPERIENCE

Holder Construction, *Office Engineer Intern – MEP Building Technology* **May 2026 - Aug. 2026**

- Managed technical documentation, requests for information (RFIs), submittals, spare parts tracking, and commissioning workflows for MEP building systems.
- Automated recurring workflows using Microsoft Power Automate.

Duke University, *Teaching Assistant – ECE 110 & EGR 105* **Aug. 2025 - May 2026**

- Supported 100+ students with Python programming, debugging, Git, and command-line workflows.

Engineers in Action (Eswatini Bridge Project), *Logistics and Media Manager* **Jan-Aug. 2025**

- Designed bridge components in AutoCAD and supported on-site construction.

Boolean Girl, *Instructor & AI Curriculum Lead* **Summer 2023, 2024**

- Developed AI and programming curricula and taught Python, Scratch, and micro:bit.

PBS, *Intern* **Summer 2022**

- Tested and documented functionality of legacy web pages based on user requirements.

PROJECTS

Book Club Web Application — Python, Django, HTML/CSS **Aug. 2024 - Feb. 2025**

- Built a full-stack application with CRUD functionality, scheduling, voting, and SMTP email delivery.

Markov Model Text Generator — Python, Machine Learning **Fall 2025**

- Preprocessed text data and implemented a probabilistic Markov model for text generation.
- Used Python to analyze model behavior and generate text based on learned transition probabilities.

International Space Station Deorbit Algorithm — Python, Matplotlib, Git **Fall 2024**

- Built a physics-based simulation to model orbital behavior and visualize results in an Agile team.

Thermal Sensor Robot — Arduino (C++), Sensors **Spring 2025**

- Built an autonomous robot integrating multiple sensors with C++ control logic.